



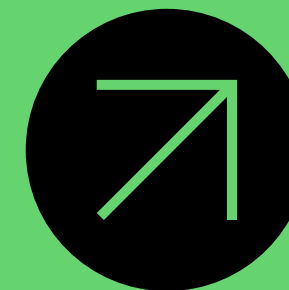
Winter 2026



STA 220



UNCOVERING GLOBAL PATTERNS THROUGH SPOTIFY DATA

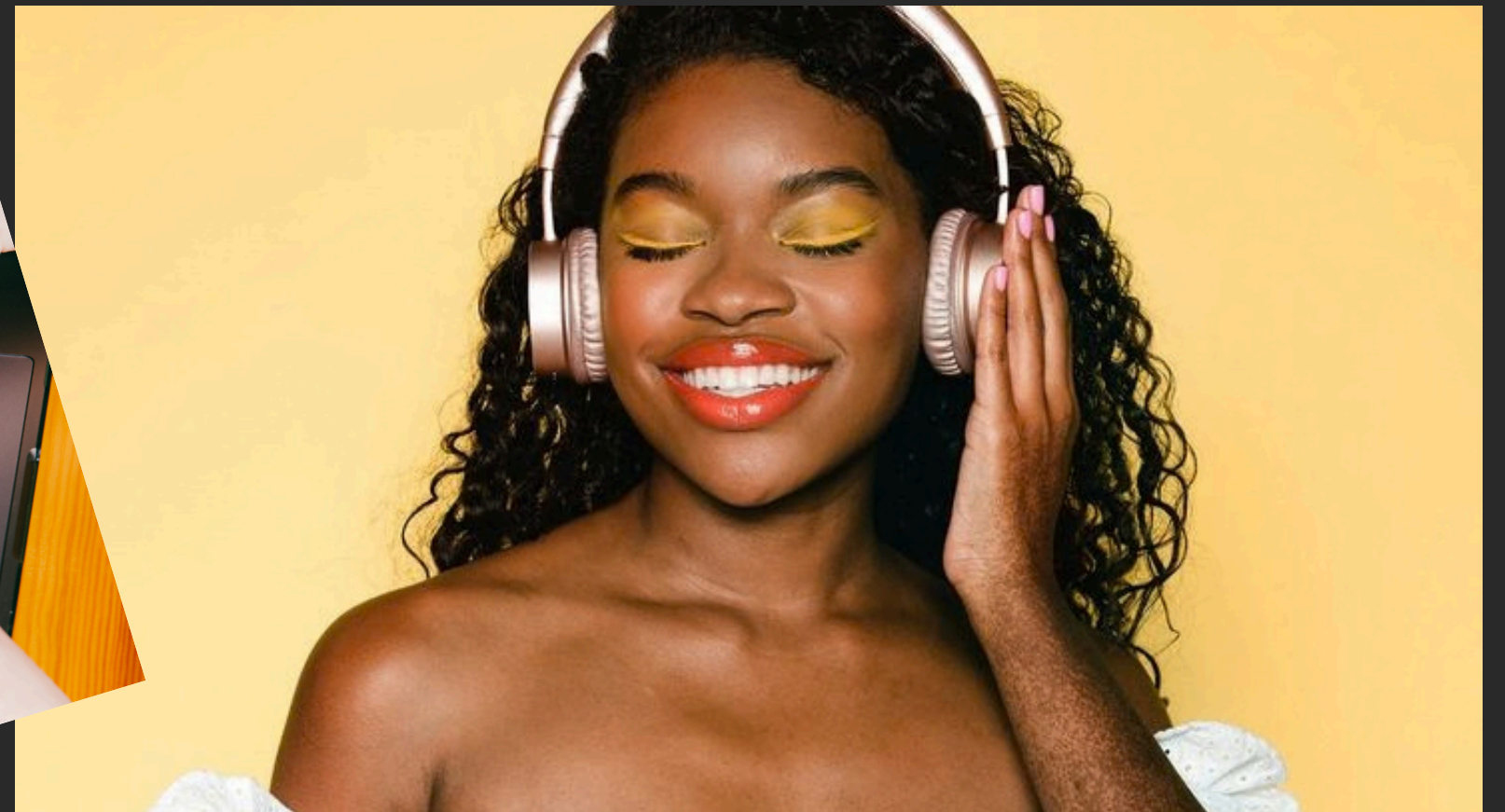


CHI-CHUN CHEN
DANNY KUEI
LIANG-YIN TAO

AGENDA

- INTRODUCTION
- DATA ACQUISITION
- ***GLOBAL STREAMING TRENDS***
- ***SONG TREND ANALYSIS***
- ***LYRICAL CONTENT MINING***
- DISCUSSION

INTRODUCTION.



Spotify Annual Wrapped...?



RESEARCH QUESTIONS

- Spotify is a major global platform spanning 184 markets
- Streams as a measure of artist popularity

1. What are Spotify user listening preferences and how do they differ amongst countries?

- **Are these preferences representative of a globalized music market or is success dominated by select few countries or genres**

2. What are the trends and characteristics of the most streamed spotify songs?

PROJECT OBJECTIVES AND METHODS

- **Global Streaming Trends:** Compare genre, users, music consumption between countries maps and visualizations.
- **Song Trend Analysis:** To identify and compare songs characteristics and trends within specific nations.
- **Semantic & Lyric Exploration:** To perform text mining on lyrics to uncover recurring themes, sentiments, and linguistic patterns that define modern popular music.



DATA ACQUISITION

DATA ACQUISITION

Web-Scraping

- **Kworb: Spotify Charts Weekly Top Songs**

Official API

- **Spotify for Developers: Song Detailed Info**
- **MusicBrainz: Database for artist info**

Charts

ITUNES	WORLDWIDE	ARTISTS	CHARTS	BETTER LOVE
RADIO	SPOTIFY	YOUTUBE	TRENDING	HOME

Countries	Artists	Listeners	Top Lists
-----------	---------	-----------	-----------

Spotify Charts | Aggregates of all daily and weekly charts provided by [Spotify](#).

Global	Daily (Totals) Weekly (Totals)
United States	Daily (Totals) Weekly (Totals)
United Kingdom	Daily (Totals) Weekly (Totals)
Andorra	Daily (Totals) Weekly (Totals)
Argentina	Daily (Totals) Weekly (Totals)
Australia	Daily (Totals) Weekly (Totals)
Austria	Daily (Totals) Weekly (Totals)
Belarus	Daily (Totals) Weekly (Totals)
Belgium	Daily (Totals) Weekly (Totals)
Bolivia	Daily (Totals) Weekly (Totals)
Brazil	Daily (Totals) Weekly (Totals)
Bulgaria	Daily (Totals) Weekly (Totals)
Canada	Daily (Totals) Weekly (Totals)
Chile	Daily (Totals) Weekly (Totals)
Colombia	Daily (Totals) Weekly (Totals)
Costa Rica	Daily (Totals) Weekly (Totals)
Cyprus	Daily (Totals) Weekly (Totals)
Czech Republic	Daily (Totals) Weekly (Totals)
Denmark	Daily (Totals) Weekly (Totals)
Dominican Republic	Daily (Totals) Weekly (Totals)
Ecuador	Daily (Totals) Weekly (Totals)
Egypt	Daily (Totals) Weekly (Totals)

Spotify Weekly Chart - United States - 2026/03/05 | **Totals**

Pos	P+	Artist and Title	Wks	Pk (x?)	Streams	Streams+	Total
1	NEW	Bruno Mars - Risk It All	1	1 (x1)	12,884,594		12,884,594
2	+2	PinkPantheress - Stateside + Zara Larsson (w...	9	2 (x1)	11,706,010	+3,826,771	48,038,626
3	-1	Ella Langley - Choosin' Texas	20	2 (x2)	9,528,820	+637,046	127,906,543
4	+14	Bruno Mars - I Just Might	8	1 (x1)	9,391,514	+4,260,464	61,952,778
5	-4	Bad Bunny - DtMF	61	1 (x6)	9,107,715	-1,457,610	334,524,823
6	-3	Olivia Dean - Man I Need	28	2 (x4)	8,110,625	+24,652	210,454,634
7	+5	Dominic Fike - Babydoll	37	7	7,586,014	+1,655,869	124,559,393
8	-1	Don Toliver - E85	5	7	7,469,414	+406,940	36,313,057
9	=	Olivia Dean - So Easy (To Fall In Love)	23	4	7,185,247	+367,647	132,613,698
10	-4	Bad Bunny - BAILE INoLVIDABLE	61	2 (x2)	6,446,001	-827,240	252,413,876
11	-6	Bad Bunny - Tití Me Preguntó	68	2 (x1)	6,414,499	-1,121,730	333,357,738
12	-4	Bad Bunny - NUEVAYoL	59	3 (x2)	6,277,332	-741,471	225,317,780
13	NEW	Bruno Mars - Cha Cha Cha	1	13	6,057,711		6,057,711
14	-3	Djo - End of Beginning	93	1 (x2)	5,964,511	-68,748	425,974,884
15	-2	Don Toliver - Body	5	5	5,909,573	+32,213	33,580,821
16	NEW	Bruno Mars - God Was Showing Off	1	16	5,718,926		5,718,926
17	NEW	Alex Warren - FEVER DREAM	1	17	5,626,564		5,626,564
18	-1	Alex Warren - Ordinary	56	1 (x2)	5,504,835	+351,911	369,437,086
19	-9	Bad Bunny - EoO	52	4	5,343,477	-757,550	185,303,325
20	+16	Bella Kay - iloveitiloveitiloveit	3	20	5,324,572	+1,421,118	12,450,870

Artist Info

 **Drake** (Canadian rapper)
~ Person

- Overview
- Releases
- Recordings
- Works
- Events
- Relationships
- Aliases
- Tags
- Reviews
- Details
- Edit

Legal name: Aubrey Drake Graham

Wikipedia

Aubrey Drake Graham (born October 24, 1986) is a Canadian rapper, singer, and actor. He is credited with popularizing R&B sensibilities in hip-hop music through rap-singing. Drake first gained recognition as Jimmy Brooks in the CTV teen drama series *Degrassi: The Next Generation* (2001–2008) and began his music career by independently releasing the mixtapes *Room for Improvement* (2006), *Comeback Season* (2007), and *So Far Gone* (2009) before signing with Young Money Entertainment.

Drake's debut album, *Thank Me Later* (2010), debuted atop the *Billboard* 200. All of his subsequent studio albums—including *Take Care* (2011), *Nothing Was the Same* (2013), *Scorpion* (2018), *Honestly, Nevermind* (2022) and *For All the Dogs* (2023)—also reached number one in the US. His fourth album, *Views* (2016), led the *Billboard* 200 for 13 weeks and his sixth album *Certified Lover Boy* (2021) set

Show more...

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Discography

- Previous
- 1
- 2
- Next

Unspecified type

Year	Title	Artist	Rating	Releases
2010	 Best Of So Far	Drake	★★★★★	1

Album

 Play on ListenBrainz

Artist information

Type: Person
Gender: Male
Born: 1986-10-24 (39 years ago)
Born in: [Toronto, Ontario](#),  [Canada](#)
Area:  [Canada](#)
IPI code: 00529397022
ISNI code: [0000 0001 2032 246X](#)

Rating

★★★★★ (see all ratings)

Tags

Genres
[hip hop](#), [r&b](#), [contemporary r&b](#), [pop rap](#), [alternative r&b](#)
Other tags
[hip-hop](#), [trap rap](#)

[See all tags](#)

External links

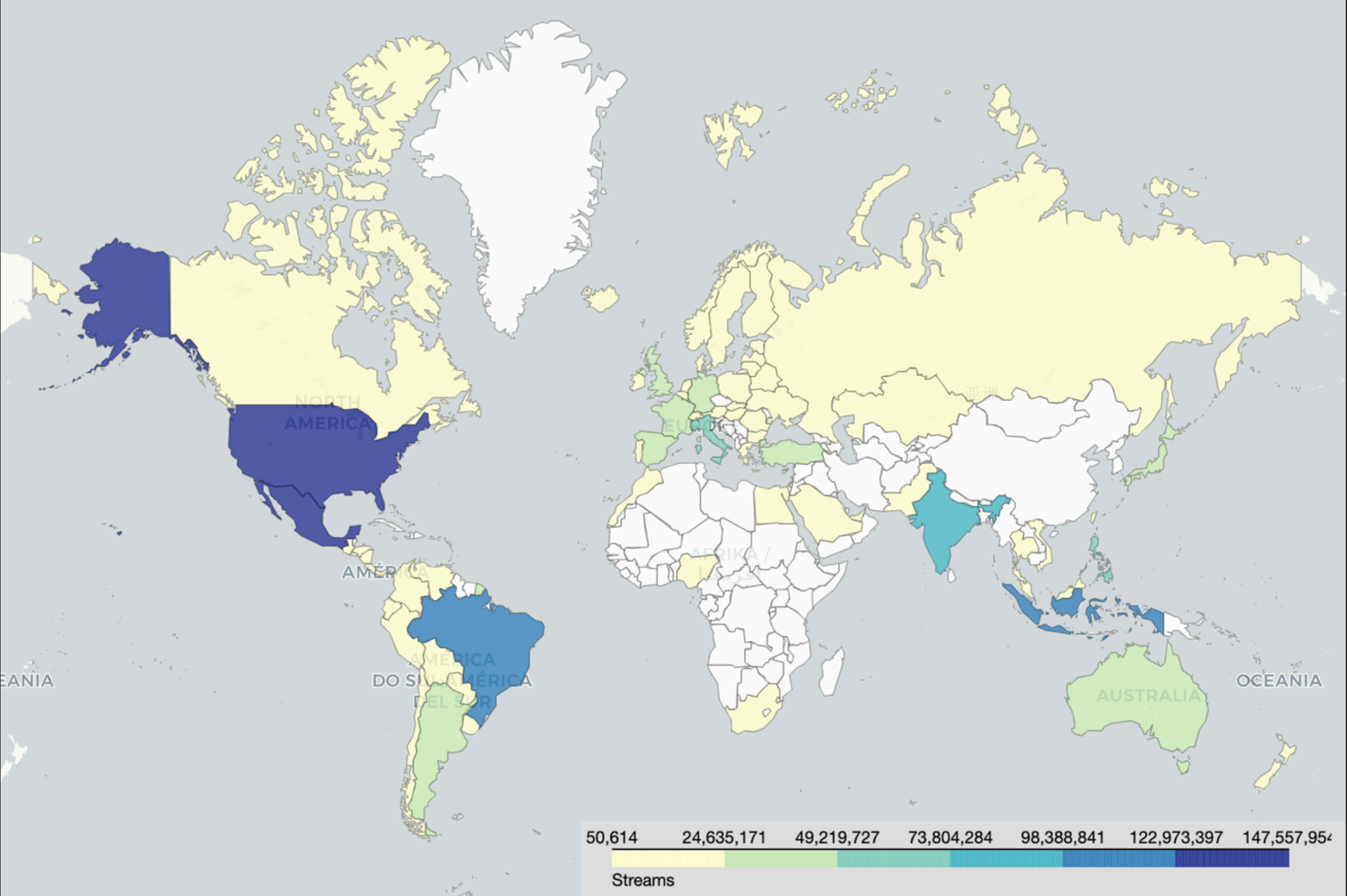
-  [Official homepage](#)
-  [@champagnepapi](#)
-  [@drake](#)
-  [facebook](#)

GLOBAL STREAMING TRENDS



Countries: Most Users Streams

Stream counts of top 20 songs for each country

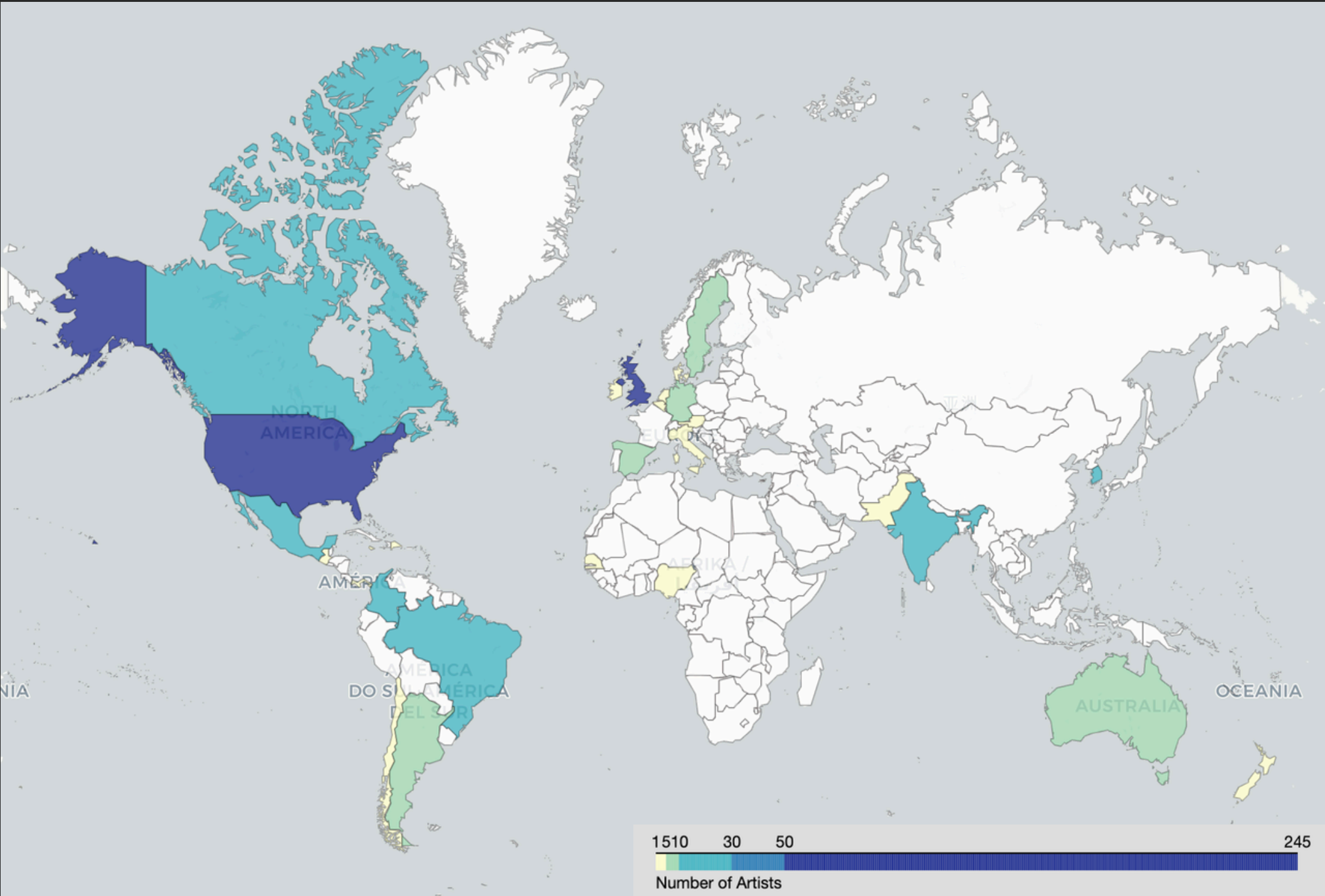


- Chart data for 76 countries

country	
0	United States
1	Mexico
2	Indonesia
3	Brazil
4	India
5	Philippines
6	Italy
7	Turkey
8	United Kingdom
9	France

Countries: Most Streamed Artists

Number of artists in top 500 global streams

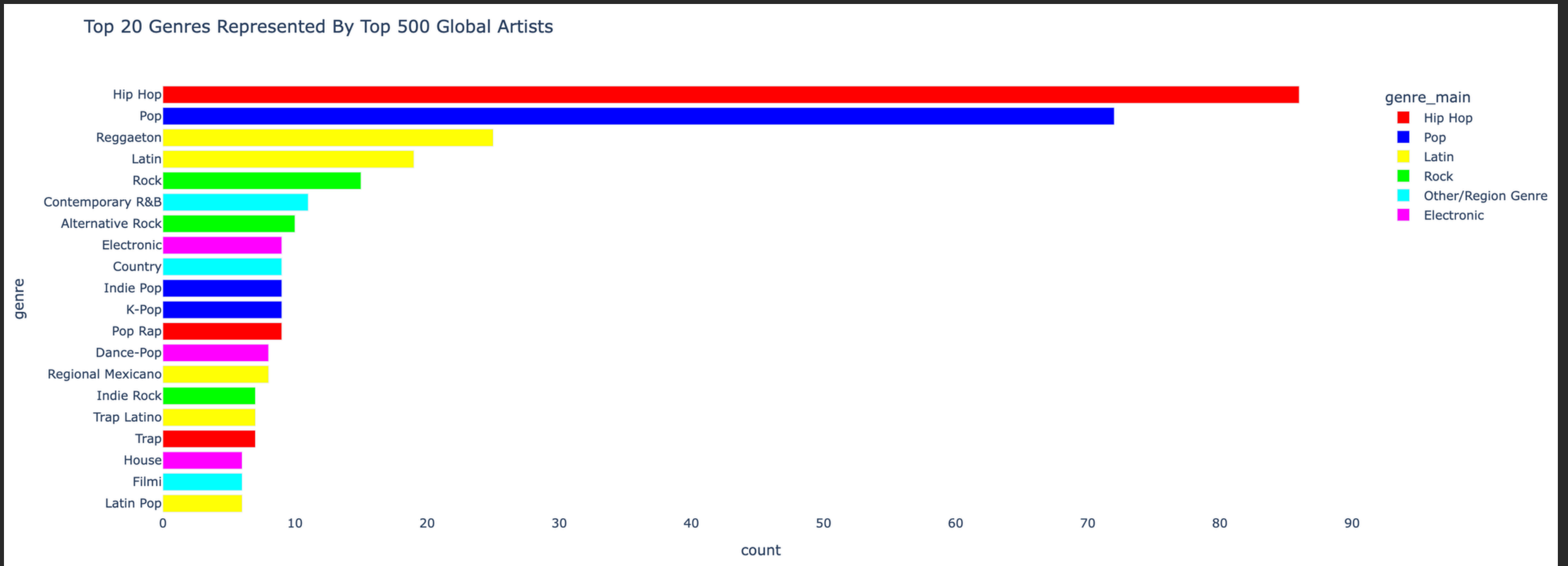


• 30 countries represented

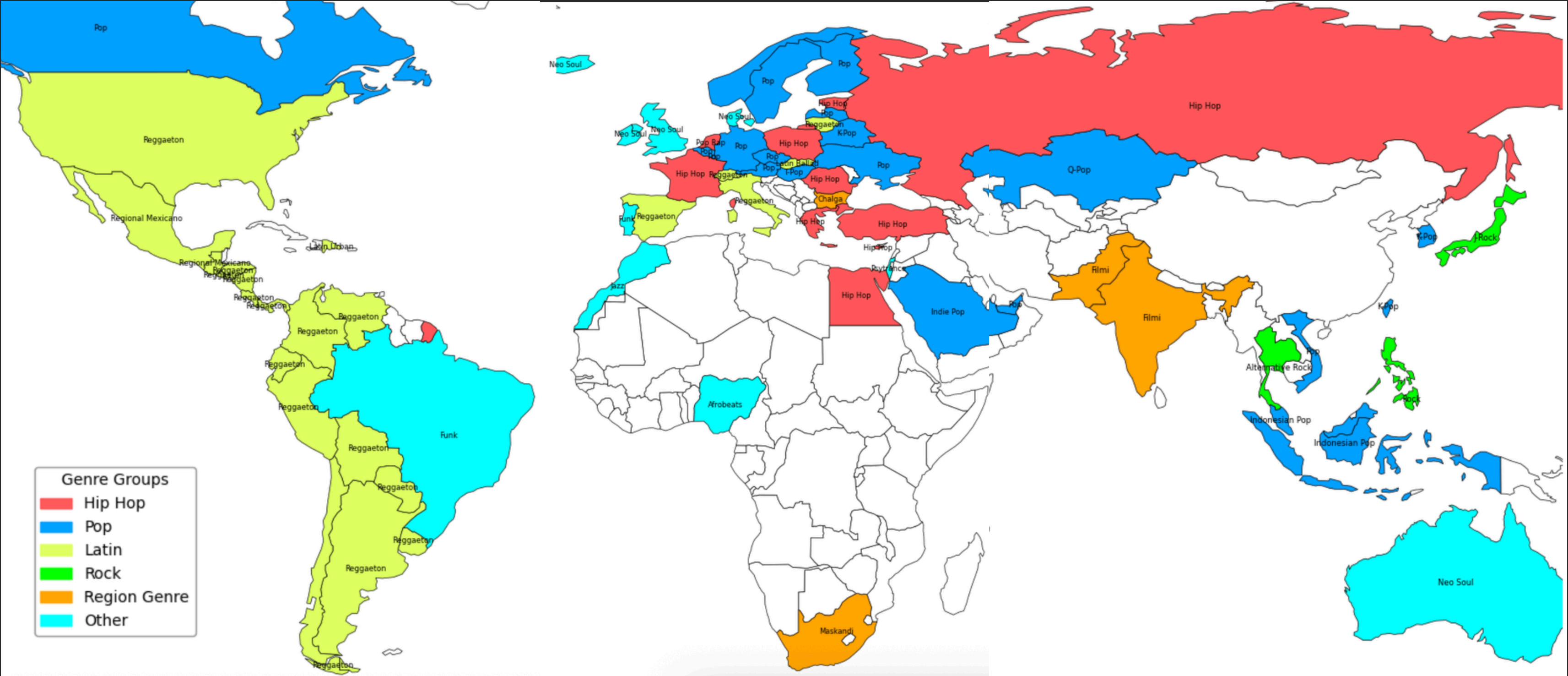
	country	count
0	United States	245
1	United Kingdom	58
2	Puerto Rico	31
3	Mexico	27
4	Canada	18
5	Colombia	14
6	India	14
7	Brazil	13
8	Korea, Republic of	11
9	Australia	9
10	Argentina	9

artist
Drake
Taylor Swift
Bad Bunny
The Weeknd
Justin Bieber
Ariana Grande
Travis Scott
Ed Sheeran
Eminem
Kanye West

Genres

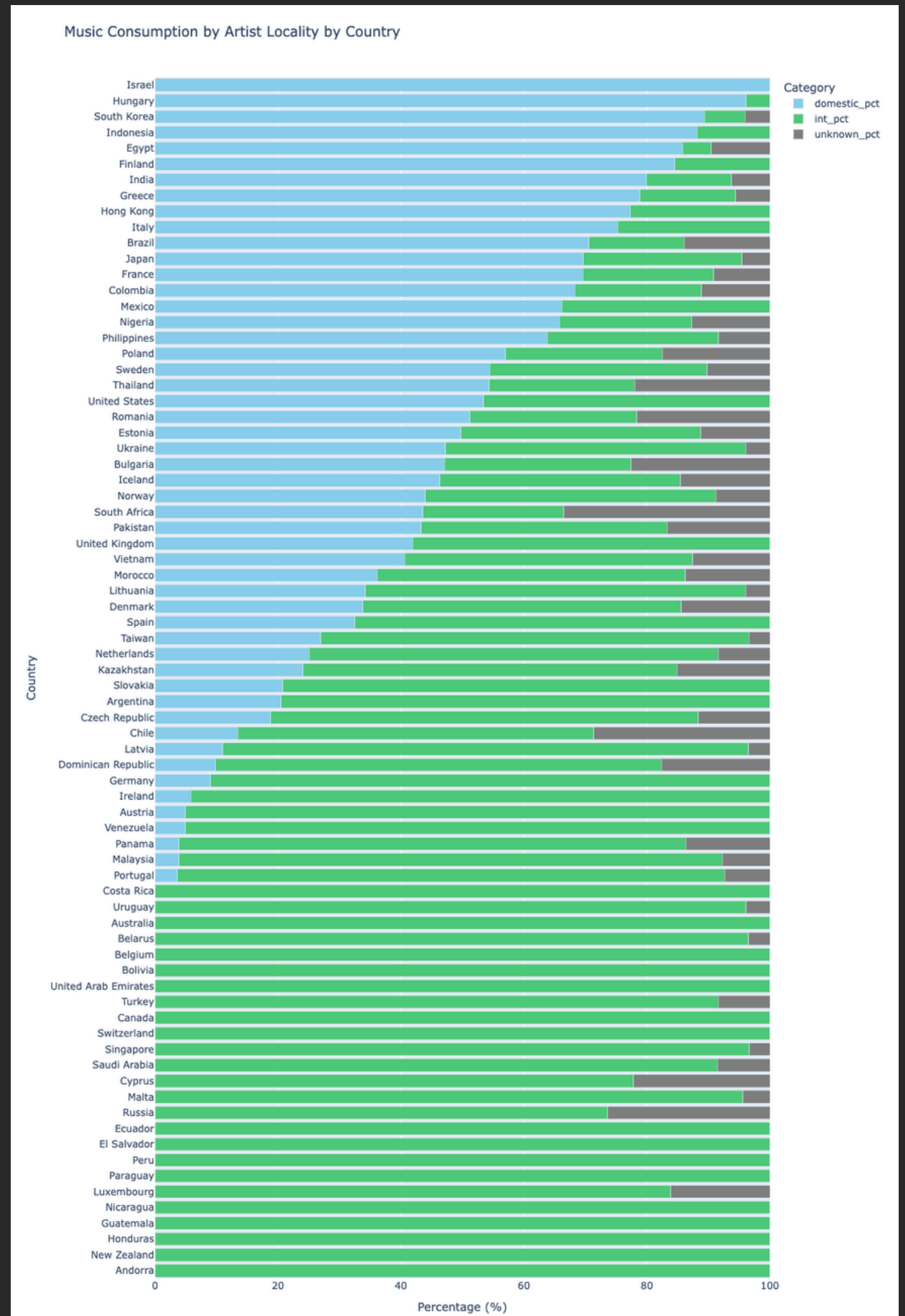


Countries by Most Streamed Genre in Top 20

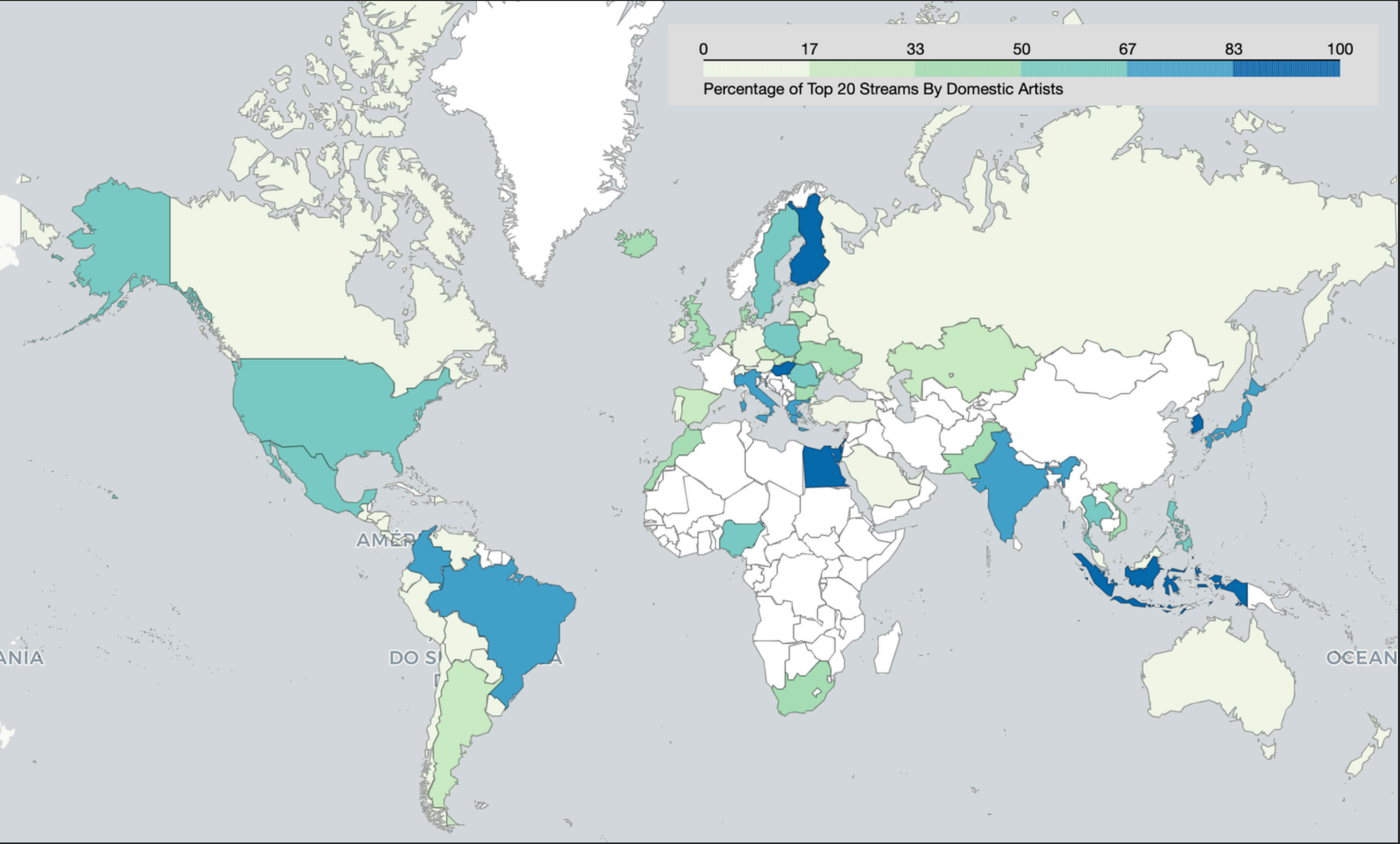


Artist Locality

- Most Percent Domestic Artists
 - Israel, Hungary, South, Korea, Indonesia, Egypt, Finland, India
- Most Percent International Artists (100%)
 - Small Latin American Countries
 - Australia, Belgium, UAE, Canada, Switzerland, New Zealand



Countries by Percentage of Artists in Top 20 Charts that are Domestic





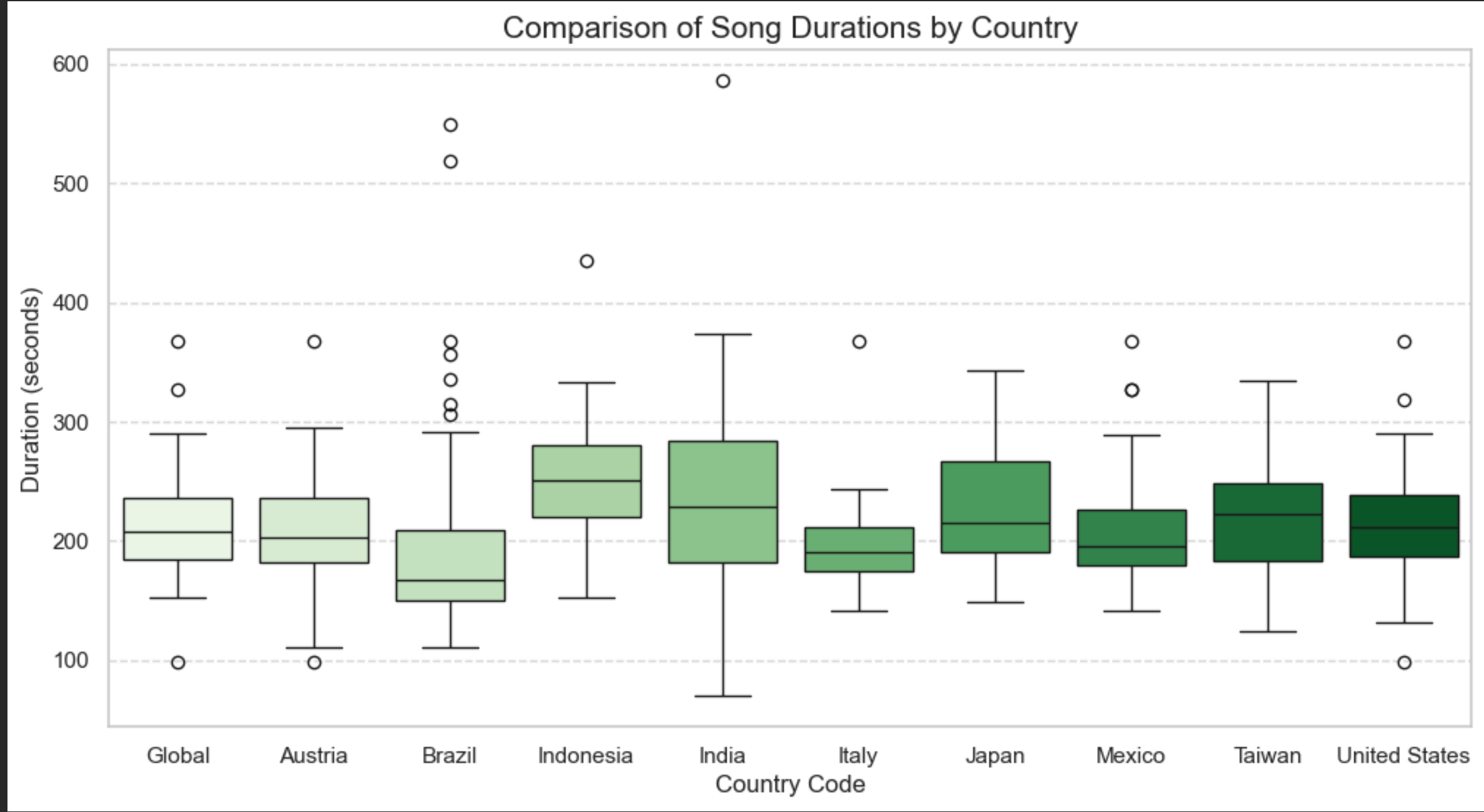
SONG TREND ANALYSIS



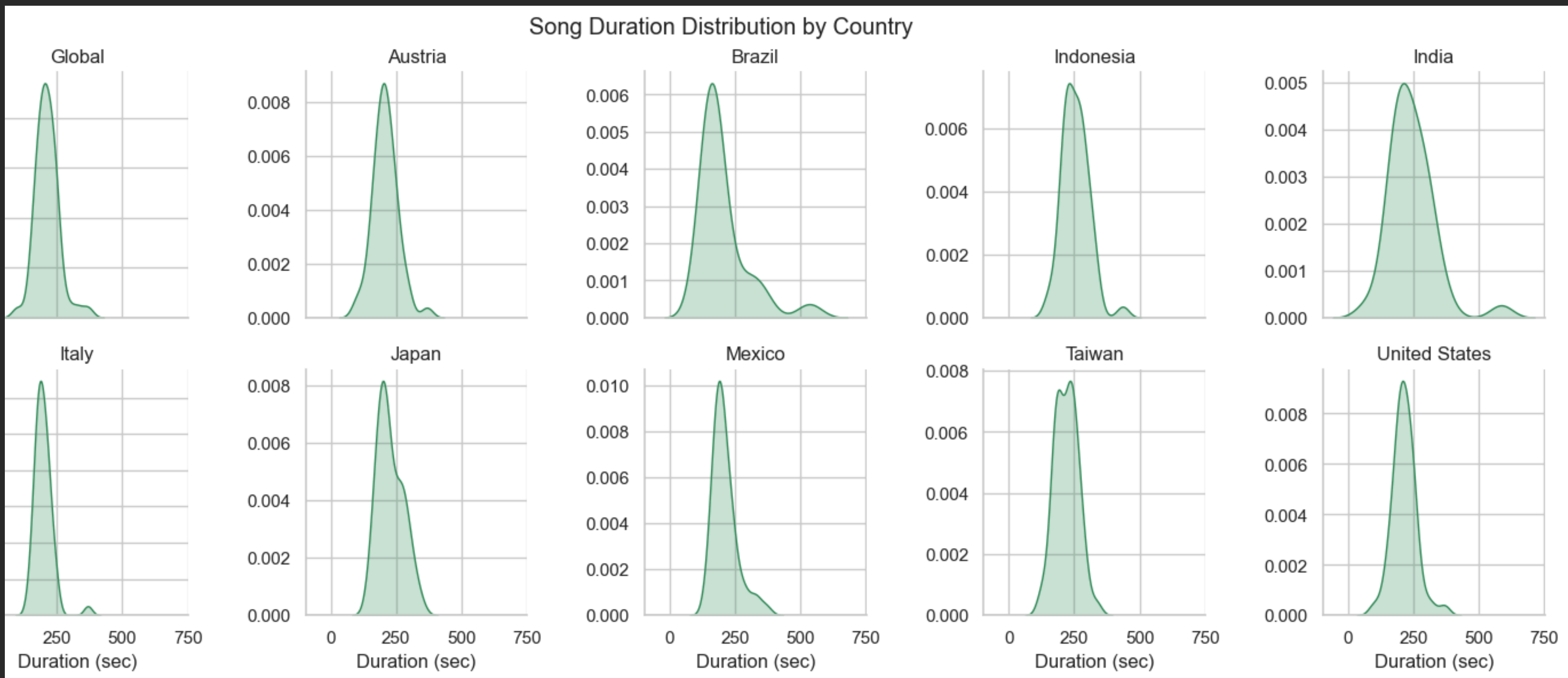
CONTENT RETRIEVED FROM API

- **Song Duration**
- **Explicit Label**
- **Song Released Date**

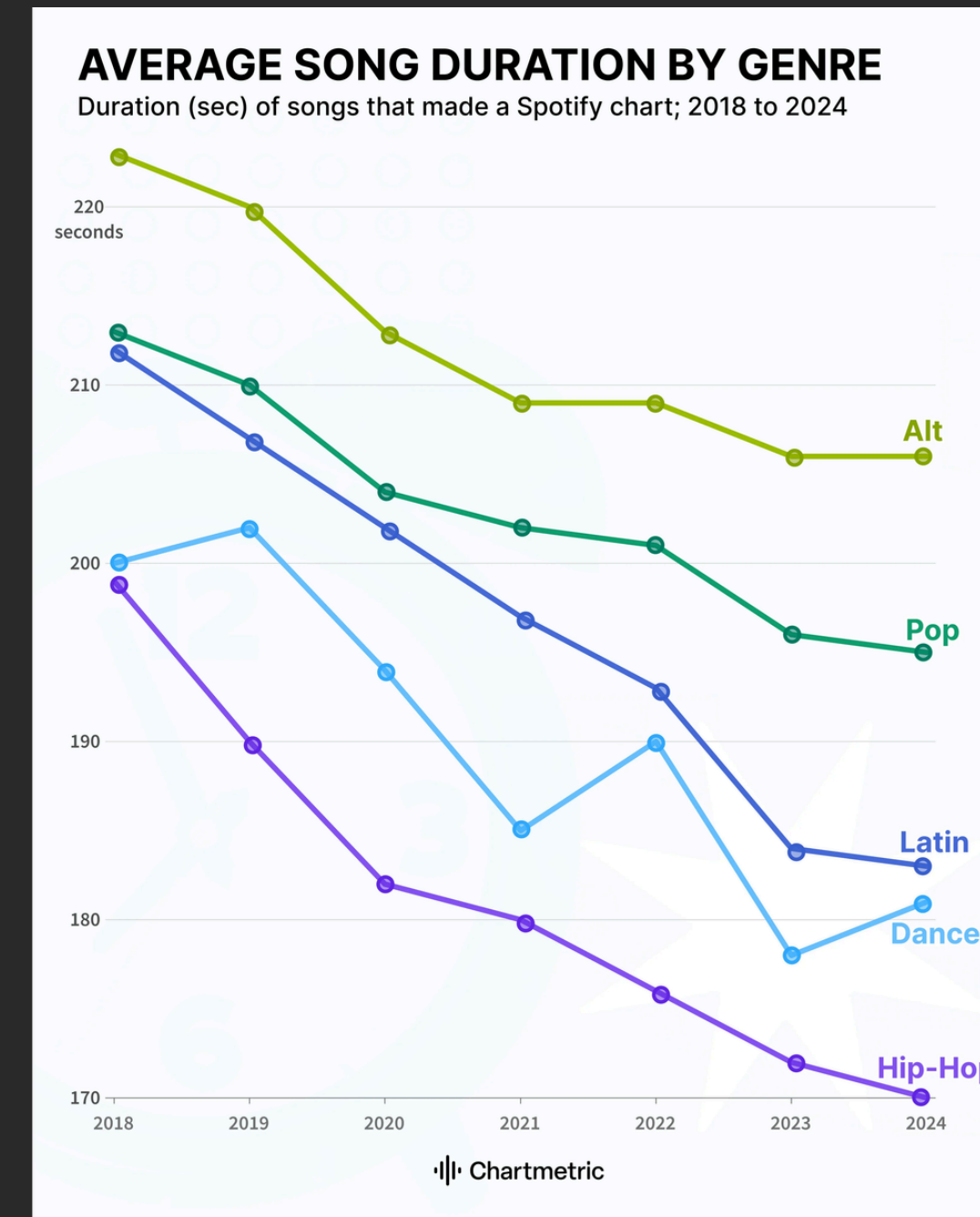
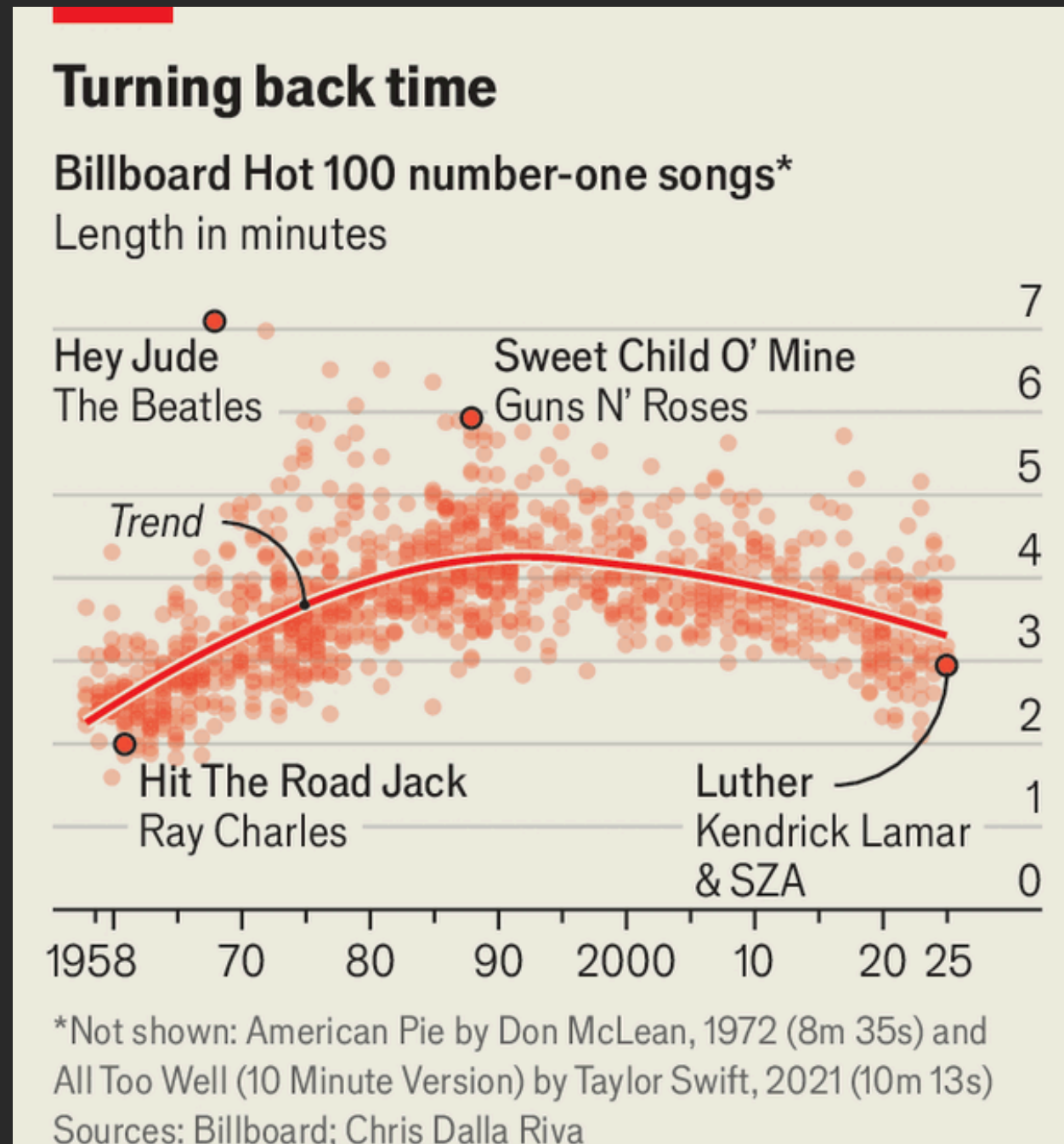
Song Durations by Countries



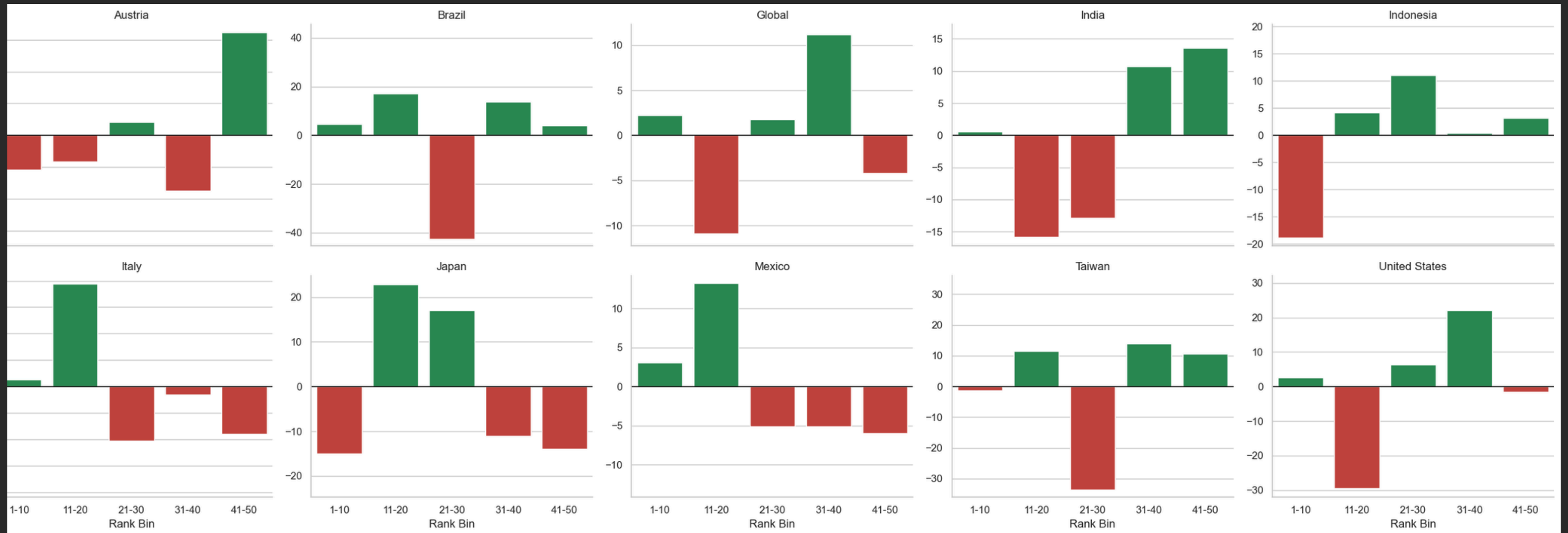
Song Durations by Countries



Do people like shorter song?



Does Song Length Affect Rank?



Spearman Correlation Test

Country	Spearman's ρ (Rho)	P-value	Significant?
Global	0.08	0.60	No
Austria	0.19	0.19	No
Brazil	-0.05	0.74	No
India	0.01	0.93	No
Indonesia	0.13	0.37	No
Italy	-0.11	0.47	No
Japan	-0.06	0.66	No
Mexico	0.04	0.79	No
Taiwan	0.08	0.60	No
United States	0.20	0.17	No

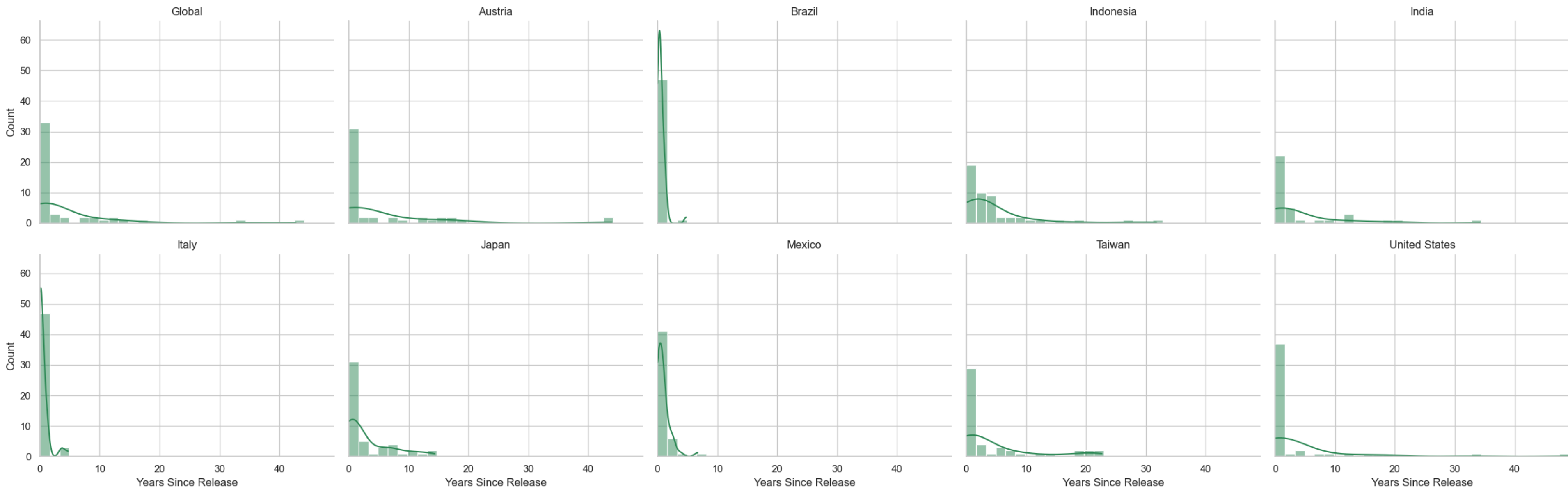
Explicit vs. Clean:

Does Genre Drive Explicit Ratios?

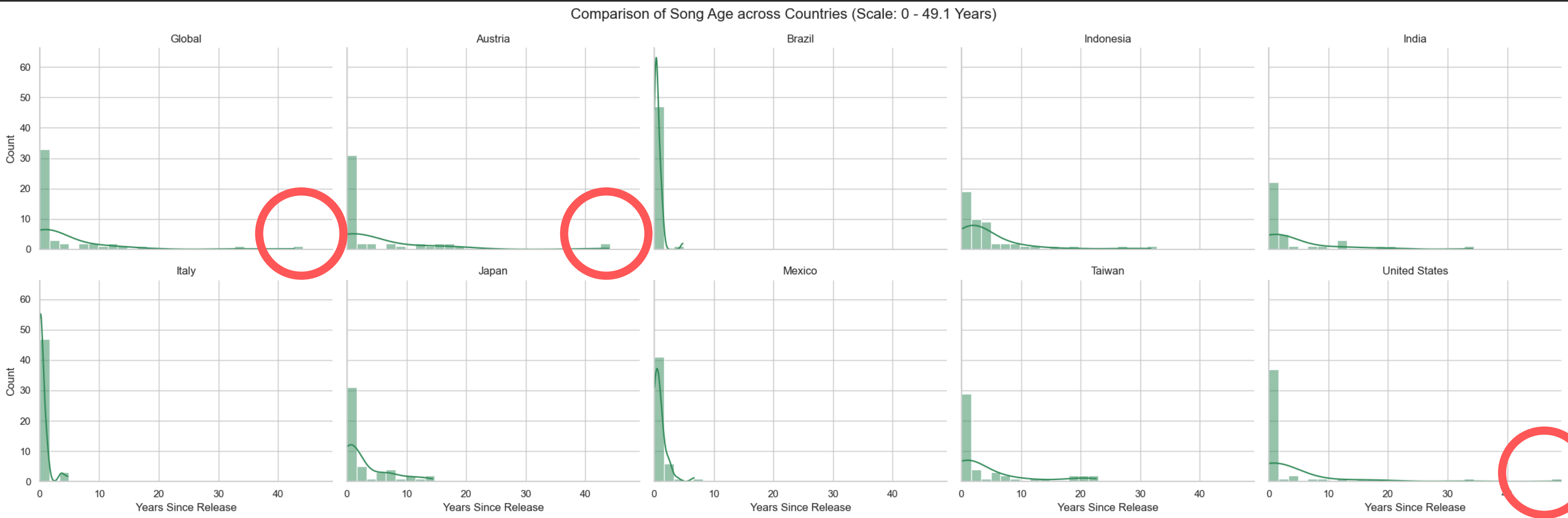
Rank	Region / Country	Explicit Count	Explicit Percentage
1	United States	16 / 50	32%
2	Global	11 / 50	22%
3	Taiwan	2 / 50	4%
4	Other Countries	0 / 50	0%

Freshness vs. Nostalgia

Comparison of Song Age across Countries (Scale: 0 - 49.1 Years)



Freshness vs. Nostalgia



Global #26 / Austria #21



Every Breath You Take
The Police • 1983

Stranger Things!



Austria #42



Africa
Toto · 1982

Where is France?



Here

Where is Brazil?



Here

Where is Africa by Toto?



Here

United States #26



Dreams
Fleetwood Mac • 1977



Ocean Spray Gives Viral Skateboarding TikTok Star a
New Truck | Eater

[Visit >](#)

Winter 2026



STA 220

**Can you guess which artist
has most songs on the Top 50
chart in your country?**



Global

#1 Bad Bunny
#2 Bruno Mars
#3 sombr

Austria

#1 Bad Bunny
#2 sombr
#3 Dominic Fike

Italy

#1 Bad Bunny
#2 Geolier
#3 Kid Yugi

United States

#1 Bad Bunny
#2 Don Toliver
#3 Ella Langley

Indonesia

#1 Hindia
#2 Nadhif Basalamah
#3 Raim Laode

Brazil

#1 Bad Bunny
#2 Danilo e Davi
#3 Henrique & Juliano

Japan

#1 Mrs. GREEN APPLE
#2 back number
#3 HANA

Taiwan

#1 Crowd Lu
#2 EggPlantEgg
#3 BABYMONSTER

India

#1 Anuv Jain
#2 Karan Aujla, Ikky
#3 (multiple)

Mexico

#1 Peso Pluma, Tito Double P
#2 Bad Bunny
#3 Fuerza Regida

LYRICAL CONTENT

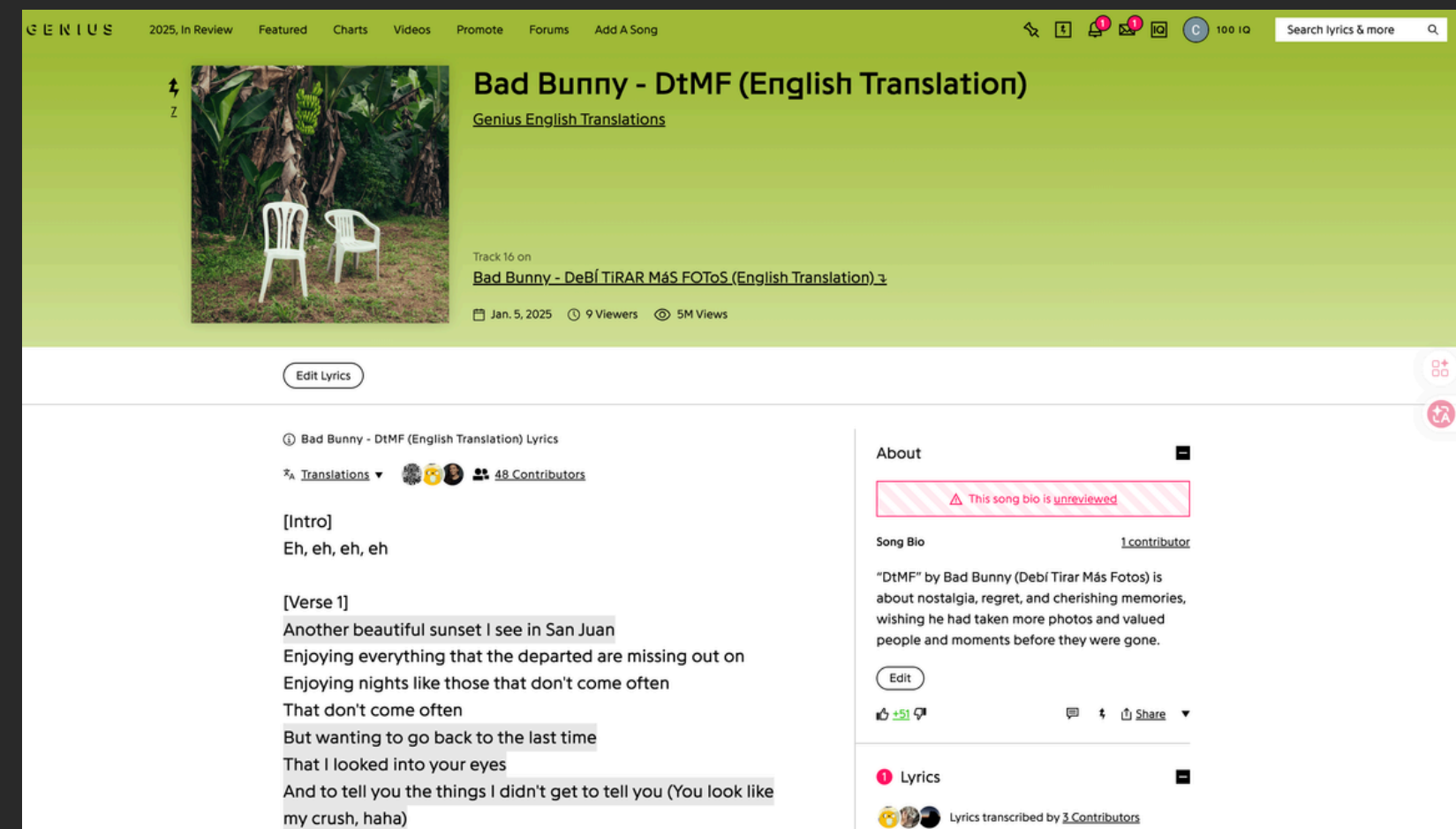


MINING



CONTENT RETRIEVED FROM API

- **Dataset: Spotify US Top 200 songs**
- **Collected lyrics for 198 songs**
- **Obtained from Genius API**



The screenshot displays the Genius website interface for the song "Bad Bunny - DtMF (English Translation)". The header includes navigation links like "2025, In Review", "Featured", "Charts", "Videos", "Promote", "Forums", and "Add A Song". A search bar is located in the top right corner. The main content area features a green background with a photo of two white plastic chairs in a tropical setting. The song title "Bad Bunny - DtMF (English Translation)" is prominently displayed, along with a link to "Genius English Translations". Below the title, it indicates the track is on "Bad Bunny - DeBí Tirar Más Fotos (English Translation)". The lyrics section shows the following text:

[Intro]
Eh, eh, eh, eh

[Verse 1]
Another beautiful sunset I see in San Juan
Enjoying everything that the departed are missing out on
Enjoying nights like those that don't come often
That don't come often
But wanting to go back to the last time
That I looked into your eyes
And to tell you the things I didn't get to tell you (You look like my crush, haha)

The right sidebar contains an "About" section with a warning that the song bio is unreviewed, a "Song Bio" section with a contributor link, and a "Lyrics" section with a contributor link.

Lyrics Analysis Framework

Lyrics



Text Preprocessing

Remove stop words...



1. Topic Modeling (LDA)

Discover hidden themes in lyrics

- Topic modeling
- Word distribution per topic
- Identify recurring themes

2. Sentiment Analysis

Emotional tone

- VADER sentiment analysis
- Positive vs negative tone

3. Keyword Analysis

Language patterns

- Frequent keywords and phrases

Text Preprocessing

- Convert text to lowercase
- Remove punctuation and special characters
- Remove section labels (e.g., [Chorus], [Verse])
- Remove stopwords
- Tokenize lyrics into words

[Intro]

Eh, eh, eh, eh

[Verse 1]

Another beautiful sunset I see in San Juan
Enjoying everything that the departed are
missing out on
Enjoying nights like those that don't come often
.....

[Chorus]

I should've taken more pictures when I had you
I should've given you more kisses and hugs
whenever I could
Ayy, I hope my people never move away
.....

[Interlude]

Guys, I love you so much, I really do
Thank you for being here, truly
It's very important to me that you're here
.....

[Verse 3]

Now Bernie has a baby, and Jan has a girl
We're no longer about the flashy stuff and chains
We're here for the things that are truly worth it
.....

Eh, eh, eh, eh

Another beautiful sunset I see in San Juan
Enjoying everything that the departed are
missing out on
Enjoying nights like those that don't come often
.....

I should've taken more pictures when I had you
I should've given you more kisses and hugs
whenever I could
Ayy, I hope my people never move away
.....

Hey, today I'll be with Grandpa all day, playing
dominoes
If he asks if I still think about you, I'll say no
That my time being close to you is over now
.....

Guys, I love you so much, I really do
Thank you for being here, truly
It's very important to me that you're here
.....

Now Bernie has a baby, and Jan has a girl
We're no longer about the flashy stuff and chains
We're here for the things that are truly worth it
.....

I should've taken more pictures when I had you

LDA Topic Modeling

LDA assumes that:

- each song is a mixture of topics
- each topic is a distribution of words

Steps:

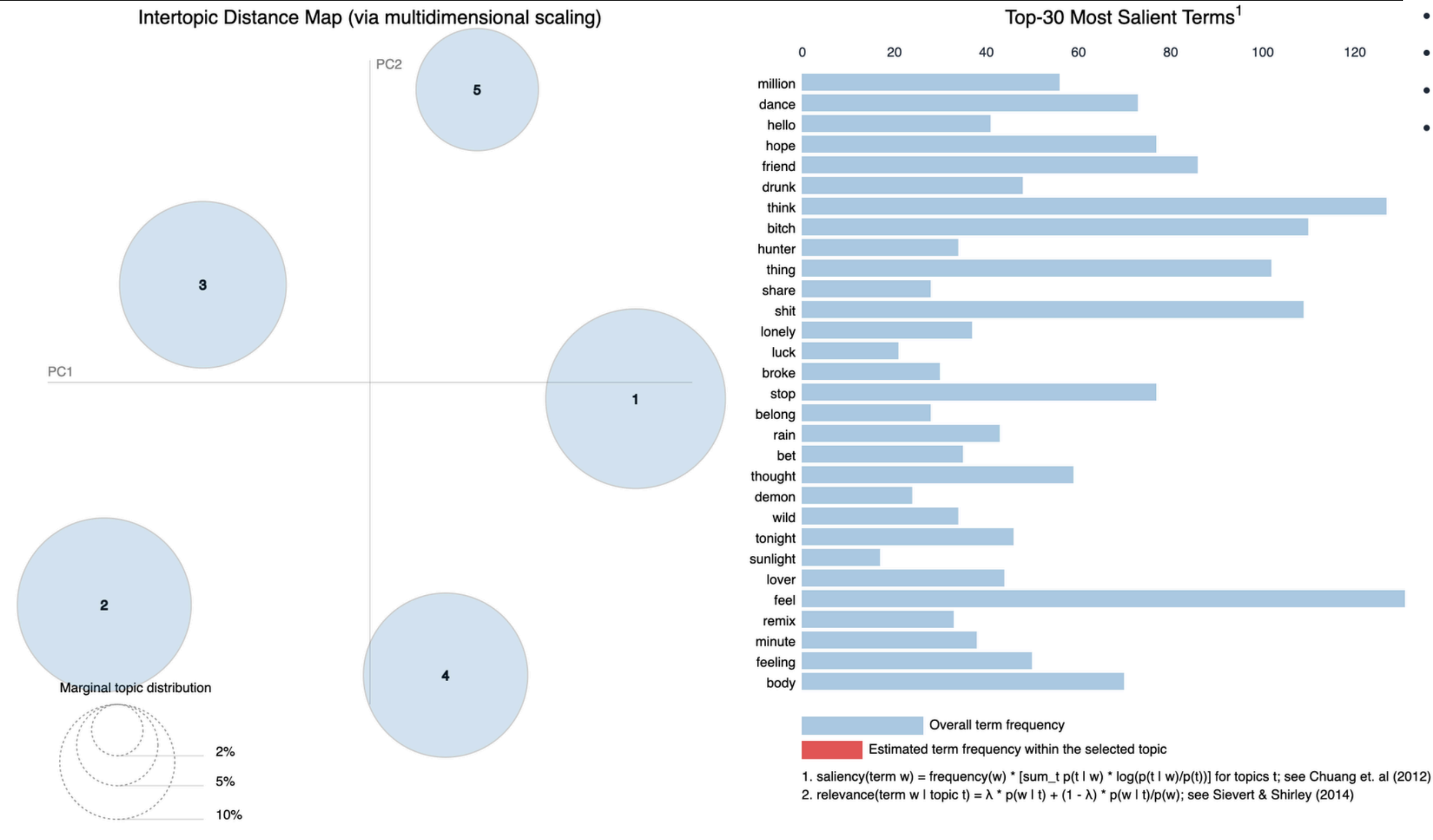
1. Tokenize lyrics
2. Build document-term matrix
3. Train LDA model
4. Extract top words per topic

LDA Results

Topic	Interpreted Theme	Representative Words
0	Upbeat/ Nightlife	dance, night, music, club, crazy, break
1	Vice/ Explicit Relationships	drunk, sex, kiss, house, fuck, night
2	Romantic Relationships/ Heartbreak	heart, lover, hurt, mind, forever, feel
3	Sad/ Loneliness	lonely, hope, broke, feeling, heartless, ocean
4	Rap/ Hip-hop	million, remix, savage, king, young, smoke

LDA Results

- Topic Label Mapping
- **Topic 1:** Romantic emotion / separation
 - **Topic 2:** Summer / music / nighttime mood
 - **Topic 3:** Nightlife / city / club atmosphere
 - **Topic 4:** Drinking / memories / emotional reflection
 - **Topic 5:** Heartbreak / friendship / sadness



Sentiment Analysis

Goal: Analyze the emotional tone of lyrics in the Top 200 songs.

Method:

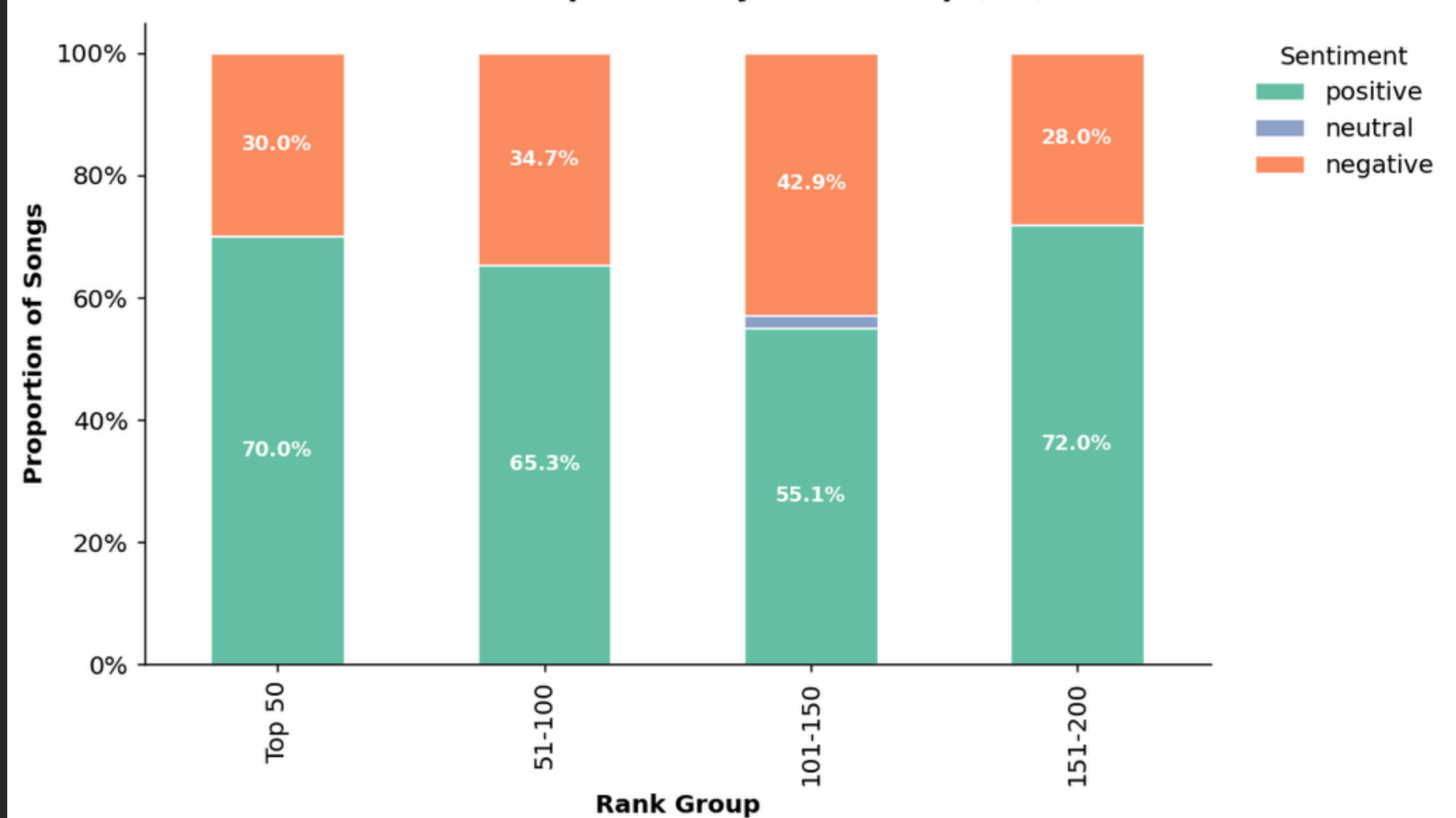
- **We applied VADER sentiment analysis.**
- **VADER assigns sentiment scores based on a sentiment lexicon.**
- **Each lyric receives:**
 - **positive**
 - **neutral**
 - **negative**
 - **compound score (-1 to 1)**

Interpretation

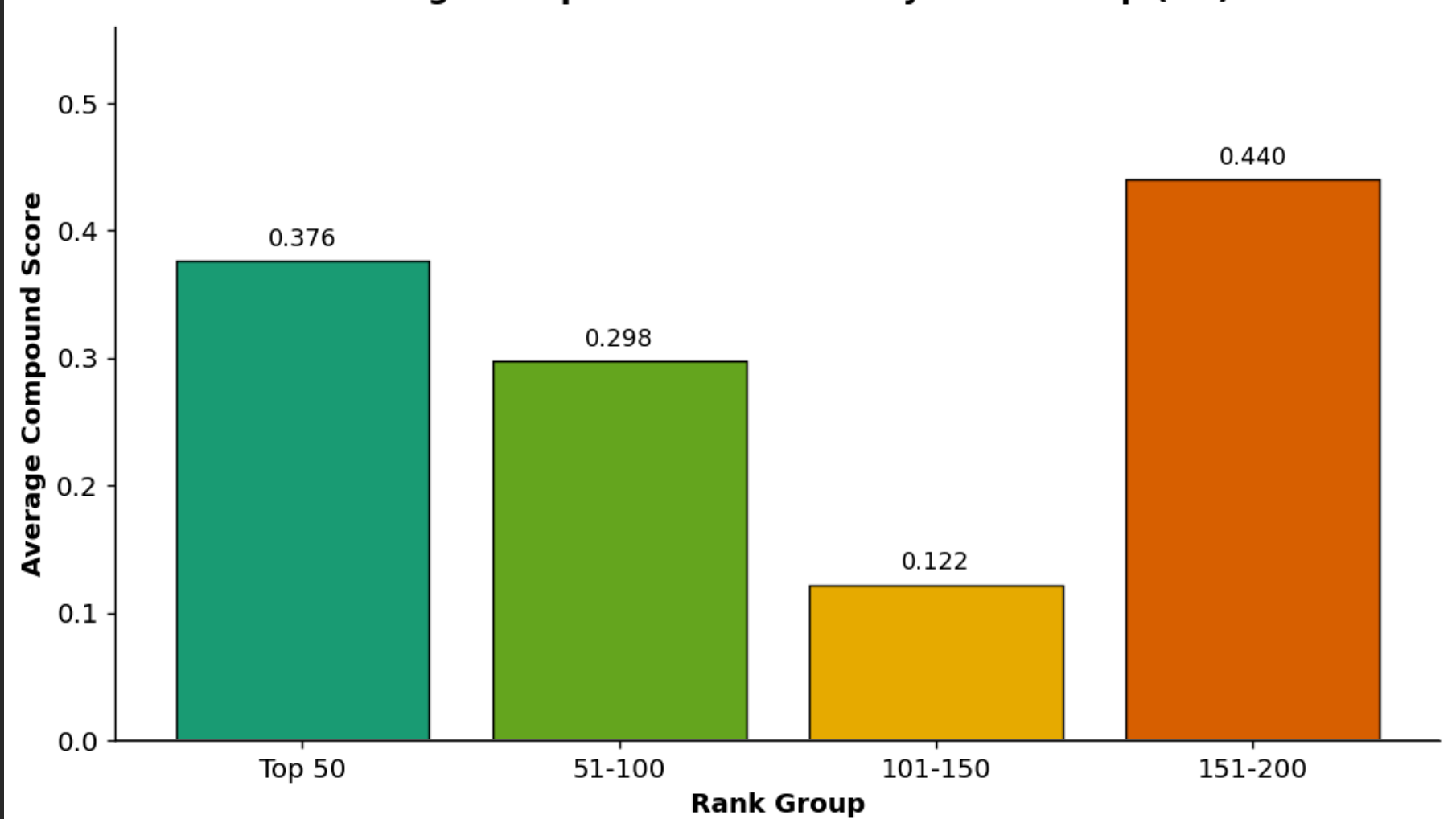
- **compound > 0.05 → positive**
- **compound < -0.05 → negative**

Sentiment Analysis Result

Sentiment Composition by Rank Group (US)



Average Compound Sentiment by Rank Group (US)



Keyword / Phrase Analysis

We analyzed common unigrams and n-grams in lyrics.

Common themes include:

- **love and relationships**
- **nightlife and partying**
- **emotional expression**

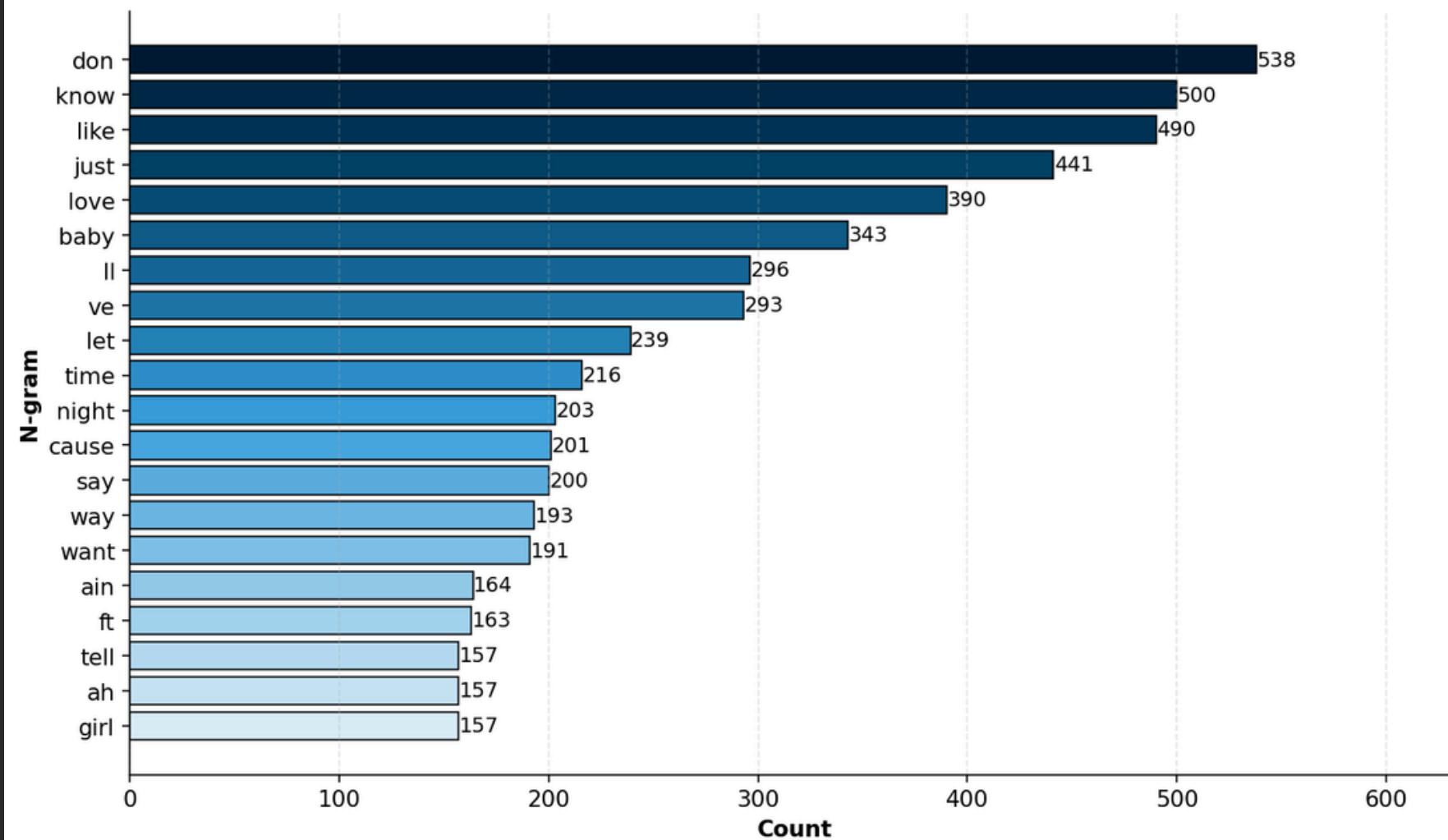
Example frequent phrases:

- **"feel like"**
- **"right now"**
- **"love you"**

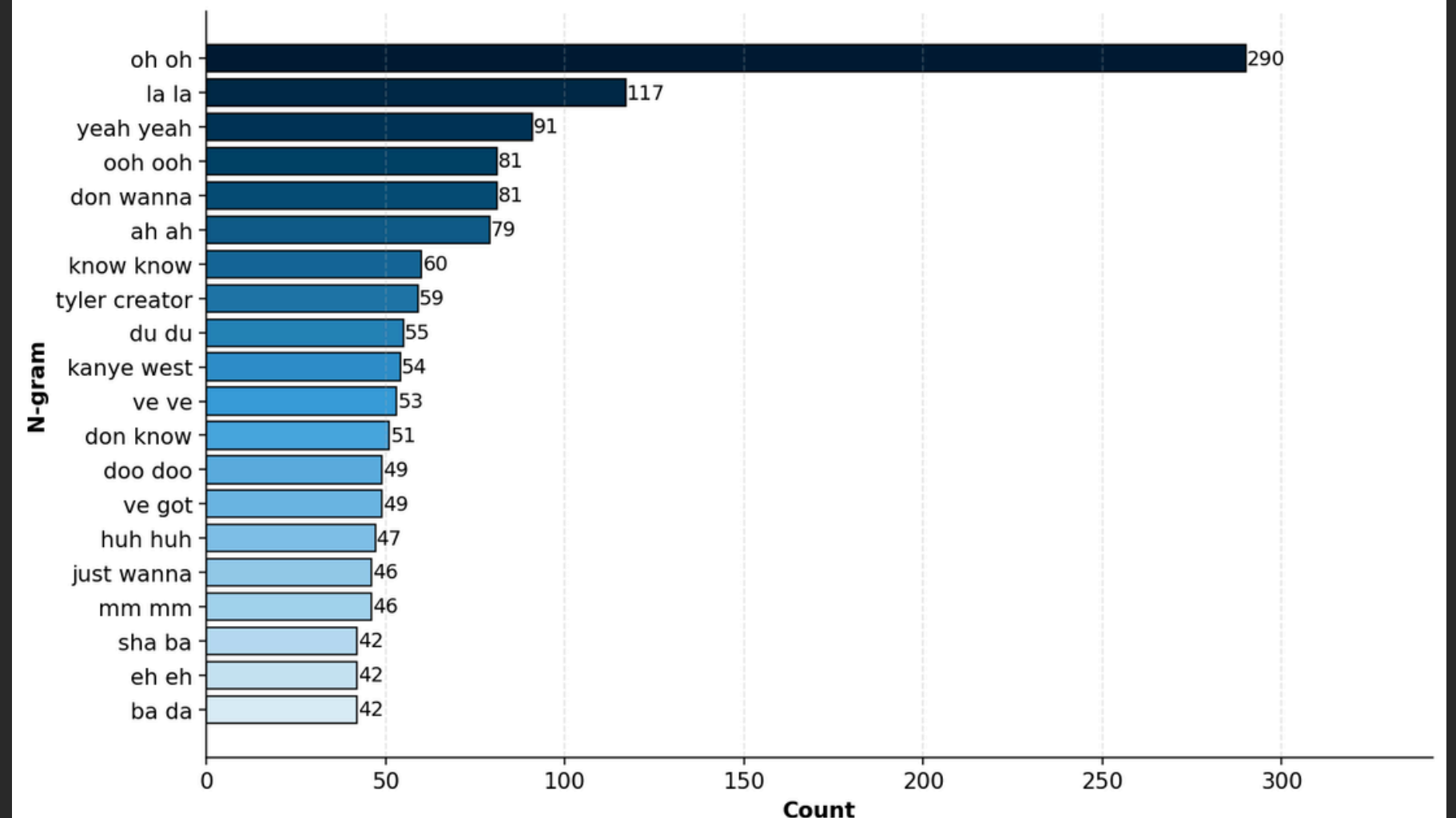
These patterns suggest recurring emotional and social themes in popular music.

Keyword / Phrase Analysis

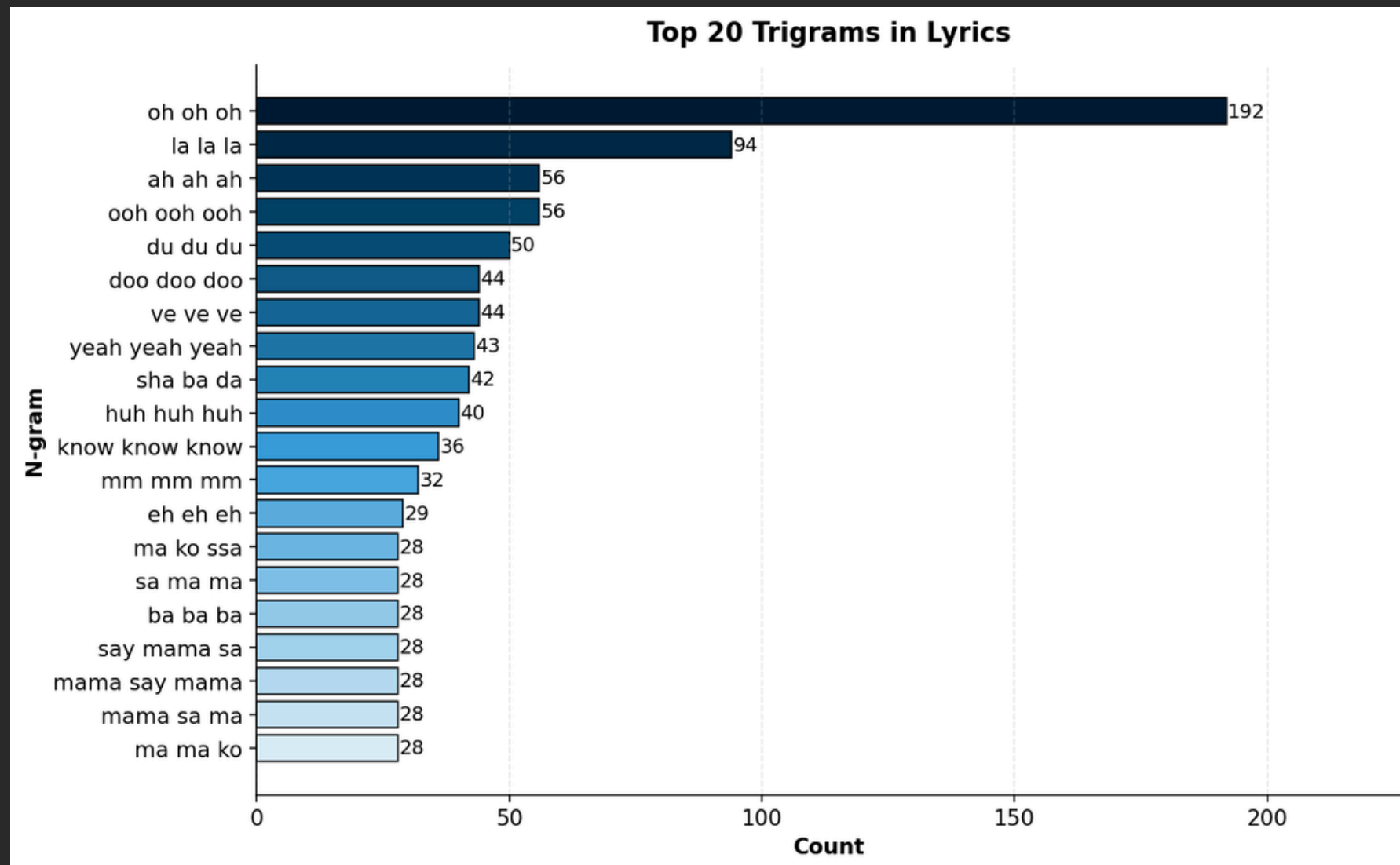
Top 20 Unigrams in Lyrics



Top 20 Bigrams in Lyrics



Keyword / Phrase Analysis



- Lyrics often focus on love, nightlife, and emotions
- Higher ranked songs tend to have more positive sentiment
- Some songs show strong semantic similarity

2026

DISCUSSION.



LIMITATIONS

- **Rate Limits**
 - **Reduced subset for charts**
- **Spotify data only not representative of population**
- **Unclean genre, lyrics and country data**

CONCLUSIONS

- **Global Trends:** Western/US influence on music market but other regions are growing to be major players
 - Hip-hop, pop are most prominent genres with reggaeton rising
- **Song Trends:** Duration distributions is not predictive of song rank and charts bias newer songs
- **Lyrics:** Reflect hip-hop and pop themes with clusters of sad, romantic, explicit, and upbeat songs

FUTURE WORK

- **Incorporate other streaming platforms (Apple Music, Youtube, etc.)**
- **Lyrical trend translation and analysis across countries**
- **Chart trends over long period of time**
- **Genre clustering**



THANK YOU

STA220 FINAL PROJECT

2026

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